

Essentials of Medical Pharmacology: Tripathi Principles

Pharmacology — MBBS 2nd Year

Standard Core Curriculum for Medical Undergraduates • Comprehensive Clinical Reference

Curriculum Year	MBBS 2nd Year
Medical Subject	Pharmacology
Target Audience	MBBS Undergraduates, Postgraduates, USMLE/PLAB Aspirants
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Chapter 1: Cardiovascular Pharmacology and Autonomics

RAAS Inhibitors: ACE Inhibitors, ARBs and Adverse Effect Profiles

Angiotensin-Converting Enzyme (ACE) Inhibitors (Enalapril, Lisinopril, Ramipril) competitively inhibit conversion of Angiotensin I to Angiotensin II. They decrease afterload, suppress aldosterone secretion, and dilate efferent glomerular arterioles, reducing intraglomerular pressure in diabetic nephropathy.

A landmark adverse effect of ACE inhibitors is intractable dry cough in 10-20% of patients. ACE is identical to kininase II, responsible for degrading bradykinin and substance P. Inhibition causes bradykinin accumulation in the bronchial mucosa, triggering cough.

Angiotensin Receptor Blockers (ARBs: Losartan, Telmisartan) competitively block AT1 receptors without inhibiting ACE/kininase II. Thus, ARBs do not elevate bradykinin and are the drug of choice when ACEi cough develops. Both agents can cause hyperkalemia and are strictly contraindicated in pregnancy (fetal renal dysgenesis).

Clinical Pearl: ACE inhibitor dry cough is mediated by bradykinin accumulation due to kininase II inhibition. Switching to an ARB (Losartan) eliminates cough because ARBs do not block bradykinin breakdown.

Bradykinin Level	Key Adverse Effects
Elevated (causes dry cough & angioedema)	Dry cough, angioedema, hyperkalemia, acute kidney injury in bilateral renal a
Unchanged / Normal (no cough)	Hyperkalemia, hypotension, dizziness, acute renal impairment in renal artery

Chapter 1: Cardiovascular Pharmacology and Autonomics

Beta-Adrenergic Antagonists: Receptor Selectivity and Overdose Management

Beta-blockers are classified by receptor selectivity: (1) Non-selective (Propranolol, Timolol) block beta-1 and beta-2; (2) Cardioselective beta-1 (Metoprolol, Bisoprolol, Atenolol, Esmolol) block cardiac beta-1 receptors; (3) Mixed alpha-1 and beta blockers (Carvedilol, Labetalol) produce peripheral vasodilation and cardiac blockade.

Cardioselective agents reduce heart rate, myocardial contractility, and renin release. However, selectivity is lost at high doses, risking severe bronchospasm in asthma/COPD due to beta-2 blockade.

Beta-blocker overdose causes severe bradycardia, cardiogenic shock, heart block, and hypoglycemia. The specific antidote is intravenous Glucagon. Glucagon binds myocardial glucagon receptors coupled to Gs, bypassing blocked beta receptors to increase intracellular cAMP, restoring inotropy and chronotropy.

Clinical Pearl: Glucagon is the first-line antidote for beta-blocker overdose because it activates adenylate cyclase and cAMP independent of beta-adrenergic receptors.

	Cardioselectivity	Special Clinical
	Cardioselective	First-line in post
	Non-selective	Thyroid storm, e
	Mixed antagonist	Carvedilol in ch
	Ultra-short acting (t1/2 ~9m)	Intraoperative S

Chapter 2: Antimicrobial Chemotherapy and Resistance

Antimicrobial Therapy for Central Nervous System Infections: Ceftriaxone, Vancomycin, Ampicillin, and Chemoprophylaxis

Antimicrobial treatment of Acute Bacterial Meningitis demands high-dose, bactericidal intravenous antibiotics capable of crossing the inflamed blood-brain barrier. The empiric gold-standard regimen for community-acquired bacterial meningitis in immunocompetent adults (<50 years) is Ceftriaxone (2g IV q12h) PLUS Vancomycin (15-20 mg/kg IV q8-12h).

Ceftriaxone provides rapid bactericidal CSF penetration against *N. meningitidis*, *H. influenzae*, and penicillin-sensitive *S. pneumoniae*. Vancomycin is mandatory to ensure coverage against penicillin-resistant *Streptococcus pneumoniae* (PRSP). In patients >50 years, neonates, or immunocompromised individuals, intravenous Ampicillin (2g IV q4h) must be added empirically to cover *Listeria monocytogenes*, which is inherently resistant to all cephalosporins.

Adjunctive Dexamethasone (10 mg IV q6h for 4 days) significantly reduces sensorineural hearing loss and mortality in pneumococcal and *H. influenzae* meningitis, but MUST be administered prior to or with the first antibiotic dose. Chemoprophylaxis for close contacts of *Neisseria meningitidis* is achieved using oral Rifampin (600 mg BID for 2 days), single-dose oral Ciprofloxacin (500 mg), or intramuscular Ceftriaxone (250 mg).

Clinical Pearl: Add Ampicillin to Ceftriaxone + Vancomycin in patients >50 years or immunocompromised to cover *Listeria monocytogenes* (which is resistant to all cephalosporins). Dexamethasone must precede or accompany the first antibiotic dose.

	Target Pathogens	Adjunctive / P
	<i>S. pneumoniae</i> , <i>N. meningitidis</i>	Dexamethason
	<i>S. pneumoniae</i> , <i>N. meningitidis</i> , <i>Listeria monocytogenes</i>	Dexamethason
	MRSA, <i>Pseudomonas aeruginosa</i> , Enterobacteriaceae	Strict CSF drain
	Eradication of nasopharyngeal meningococcal carriage	Droplet isolation

Chapter 2: Antimicrobial Chemotherapy and Resistance

Beta-Lactam Antibiotics, Cephalosporin Generations and Resistance Mechanisms

Beta-lactam antibiotics (Penicillins, Cephalosporins, Carbapenems, Monobactams) inhibit bacterial cell wall synthesis. The beta-lactam ring mimics D-alanyl-D-alanine, irreversibly binding and inactivating penicillin-binding proteins (PBPs/transpeptidases), blocking peptidoglycan cross-linking and provoking osmotic lysis.

Cephalosporin generations exhibit evolving antimicrobial spectra: 1st Gen (Cefazolin, Cephalexin) target Gram-positive cocci; 2nd Gen (Cefuroxime, Cefoxitin) add H. influenzae and anaerobes; 3rd Gen (Ceftriaxone, Ceftazidime) provide broad Gram-negative coverage and cross the blood-brain barrier for bacterial meningitis; 4th Gen (Cefepime) combines Gram-positive and anti-pseudomonal coverage; 5th Gen (Ceftaroline) binds mutated PBP2a to kill MRSA.

Resistance mechanisms include bacterial beta-lactamases (ESBLs, AmpC, carbapenemases like NDM-1), altered PBPs (mecA gene encoding PBP2a in MRSA), and decreased outer membrane porin permeability.

Clinical Pearl: Ceftriaxone is the primary cephalosporin for bacterial meningitis due to superb CSF penetration; Ceftaroline (5th Gen) is the only cephalosporin active against MRSA.

	Spectrum & Special Properties	Clinical Indications
	Gram-positive cocci, PEcK (Proteus, E. coli, Klebsiella)	Surgical wound infections
	Extended Gram-negative + anaerobes (Cefoxitin)	PID, abdominal infections
	Broad Gram-negative, excellent CSF penetration; Ceftazidime vs Pseudomonas	Bacterial meningitis
	Broad Gram-positive + Pseudomonas aeruginosa	Febrile neutropenia
	Binds mutated PBP2a; active against MRSA	Complicated skin infections

Chapter 2: Antimicrobial Chemotherapy and Resistance

Ribosomal Protein Synthesis Inhibitors: Macrolides, Aminoglycosides, and Toxicity Profiles

Protein synthesis inhibitors exploit differences between bacterial 70S (50S + 30S) and eukaryotic 80S ribosomes.

Macrolides (Azithromycin, Clarithromycin) bind the 23S rRNA of the 50S subunit, inhibiting peptidyl-tRNA translocation (bacteriostatic). They cover atypical respiratory pathogens (Mycoplasma, Chlamydia, Legionella). Key adverse effects include QT prolongation (torsades de pointes) and motilin-mediated GI distress.

Aminoglycosides (Gentamicin, Amikacin) bind irreversibly to the 30S subunit, causing mRNA misreading (bactericidal). They require oxygen for uptake and lack activity against anaerobes. Dose-limiting toxicities are irreversible Ototoxicity and reversible Nephrotoxicity (acute tubular necrosis).

Clinical Pearl: Remember: 30S inhibitors = Aminoglycosides, Tetracyclines; 50S inhibitors = Macrolides (Azithromycin), Chloramphenicol, Clindamycin, Linezolid.

	Mechanism	Characteristic
	Inhibits peptidyl-tRNA translocation (bacteriostatic)	QT prolongation
	Misreading of mRNA codons (bactericidal)	Ototoxicity (cochlear)
	Blocks aminoacyl-tRNA attachment to A-site	Teeth discoloration